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Validation of the Ask Suicide-Screening Questions (ASQ) with Youth in Outpatient Specialty and Primary Care Clinics

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Abstract

Objective: Validate the Ask Suicide-Screening Questions (ASQ) with youth in outpatient specialty and primary care clinics.

Method: This is a cross sectional instrument validation study assessing the validity of the ASQ with respect to the standard criterion, Suicidal Ideation Questionnaire (SIQ/SIQ Jr.). The sample

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included 515 English speaking youth ages 10–21 years old from outpatient specialty and primary care clinics. ASQ sensitivity, specificity, positive and negative predictive values (PPV/NPV), positive and negative likelihood ratios, c statistic and respective receiver operating characteristic curves were assessed.

Results: A total of 335 outpatient specialty and 180 primary care clinic participants completed the study. In outpatient specialty clinics, the ASQ showed a sensitivity of 100.0% (95% CI: 80.5 – 100.0%), specificity of 91.2% (95% CI: 87.5 – 94.1%), and NPV of 100.0% (95% CI: 98.7–100.0). In the primary care clinic, the ASQ showed a sensitivity of 100.0% (95% CI: 59.0 – 100.0%), specificity of 87.9% (95% CI: 82.0 – 92.3%), and NPV of 100.0% (95% CI: 97.7–100.0). Forty-five (13.4%) outpatient specialty clinic participants and 28 (15.6%) primary care clinic participants screened positive for suicide risk on the ASQ.

Conclusions: The ASQ is a valid screening tool for identifying youth at elevated suicide risk in outpatient clinical settings.

Keywords

screening; prevention; instrument validation

1. Introduction

Suicide is an increasingly devastating public health problem in the United States (U.S.) and the current rate is the highest it has been in decades[1,2]. In the U.S., the age-adjusted suicide rate increased by 33% from 10.5 to 14.0 per 100,000 between 1999 and 2017[3]. While suicide is a leading cause of death across age groups, suicidal thoughts and behaviors in youth are a major public health problem and warrant particular attention. In 2017, suicide was the second leading cause of death among 10–24-year-olds and 6,769 youth ages 10 to 24 died by suicide in the U.S[4]. Intervention during adolescence presents a critical opportunity to identify those at risk for suicide and potentially save lives.

Early identification of patients at risk for suicide is a key prevention strategy; however, high-risk patients often go unrecognized by healthcare providers. Death registry studies show that 80% of youth who died by suicide had contact with a healthcare provider within months of their death, and 38% of these youth (up to age 19) had visited a medical provider in the 4 weeks before a death by suicide[5–7]. Youth with medical concerns (e.g., diabetes, sports medicine, medical illness) are at increased risk for suicide and primary care is the de facto behavioral health services and care system[21–24]. Recognizing the medical setting as a critical venue to identify at-risk patients, the Joint Commission issued a Sentinel Event Alert in 2016 recommending that all medical patients be screened for suicide risk in medical settings[8].

There are several screening instruments healthcare providers can use for detecting suicide risk (e.g. Columbia Suicide Severity Rating Scale[33], the Patient Safety Screener[34] and the Suicide Behaviors Questionnaire Revised[35]), however, only the Ask Suicide-Screening Questions (ASQ)[9,36] was developed specifically for medical patients. Most tools developed for mental health screening for outpatient and primary care clinics are either costly because

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they require a great deal of time and special training or software to administer, or are intended to screen for depression[10,25,26].

The Ask Suicide-Screening Questions (ASQ)[9], a 4-item screening instrument, is a brief suicide risk screening tool developed for both medical and psychiatric patients presenting to a pediatric emergency department[9]. To make sure the tool was valid for other settings, the ASQ was then validated in inpatient medical/surgical units[36]. This study followed previous validation studies using a standard criterion measure that predicts suicide risk by assessing proxy measures of future suicidal behavior[9]. For this purpose, we chose the Suicidal Ideation Questionnaire (SIQ/SIQ Jr.), a longer instrument that measures severity of suicidal ideation (a strong predictor of suicidal behavior[27]). The SIQ/SIQ Jr. has demonstrated high reliability (SIQ: $r = 0.97$; SIQ-JR: $r = 0.94$), validity, and predictive ability[10,12] using the standard criterion the Suicidal Behavior Interview, a semi-structured clinical interview that assesses current and past suicidal behavior[8]. The concordance of the ASQ with the SIQ/SIQ Jr. in identifying individuals with elevated risk of suicide has previously been assessed for pediatric ED patients. In the pediatric emergency department ASQ development study, the ASQ was found to have high sensitivity (96.9%) and specificity (87%), and high negative predictive values for both medical (99.7%) and psychiatric patients (96.9%)[9]. A recent study has also shown the ASQ to improve the ability to predict future suicide-related behaviors (i.e., suicide-related emergency department visits and suicide deaths) beyond clinical judgment alone in both pediatric emergency department patients presenting with psychiatric problems as well as medical emergencies[11]. However, the ASQ has not been validated in outpatient non-emergency settings. It is important to assess its validity in these settings because of potential differences between emergency and non-emergency clinical settings in the context of the screening (e.g., administered by a triage nurse in a relatively stressful environment with the youth completing the screen while experiencing an urgent medical issue, versus in a more routine clinical setting and with less medical urgency) as well as potential differences in the respective samples (e.g., patients treated in emergency departments may have more acute or severe medical and psychological issues).

The primary aim of this study was to test the validity of the ASQ with youth in non-emergency settings of outpatient specialty and primary care clinics to identify individuals with elevated suicide risk, using the SIQ/SIQ Jr. as the standard criterion measure.

2. Methods

2.1 Ethics Statement

This study was approved by scientific and institutional review boards at the National Institute of Mental Health and the individual study sites. For participants younger than 18 years of age, we obtained written or verbal informed consent from the parent/guardian and written informed assent from the participant. For participants aged 18 and older, we obtained written informed consent. Participant responses were kept confidential; however, parents were notified if their minor child screened at risk for suicide. All participants who screened positive for suicide risk received an additional suicide risk assessment administered by a healthcare provider.

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2.2 Study Sites

Data came from two independent samples at two urban tertiary care academic hospitals, with study site 1 consisting of four outpatient specialty clinics at Children's Mercy Hospital Kansas City (diabetes, endocrinology, sports medicine, orthopedics) in the Midwest of the U.S. Study site 2 consisted of one primary care clinic at Boston Children's Hospital in the Northeastern U.S.

2.3 Instruments

Ask Suicide-Screening Questions (ASQ)[9].—The ASQ, is a four-item nonproprietary, self-report instrument intended to identify individuals at elevated risk of suicide. The four ASQ items assess major facets of established suicide risk factors, including suicidal ideation, perceived burdensomeness on others and past history of suicide attempt. Specifically, the four items on the ASQ are; (1) In the past few weeks, have you wished you were dead? (2) In the past few weeks, have you felt that you or your family would be better off if you were dead? (3) In the past week, have you been having thoughts about killing yourself? and (4) Have you ever tried to kill yourself? In this study, a "yes" response to any of the four questions constitutes a positive screen for suicide risk requiring further assessment from a healthcare provider, per usual care. Participants who screened positive for suicide risk were then asked an acuity question, "Are you having thoughts of killing yourself right now?". Participants who responded yes to the acuity question were considered an *acute positive* screen for suicide risk, whereas participants who responded "no" to the acuity item were considered *non-acute positive* screens.

Suicidal Ideation Questionnaire (SIQ/SIQ Jr.)[10].—While the ASQ measures perceived burdensomeness, suicidal ideation and attempt, the Suicidal Ideation Questionnaire is a self-report instrument measuring severity of suicidal ideation in adolescents. In this study, the 30-item SIQ was administered to participants 15 years of age and older and the 15-item SIQ Jr. was administered to younger participants. Each item on the SIQ begins with "I thought…", "I wondered…", "I wished…" followed by a statement indicative of suicidal ideation. Respondents indicate the frequency with which they had that particular thought within the last month on a scale from 1 ("I never had this thought") to 7 ("Almost every day"). The items range from general thoughts indicating a general wish to not be alive (e.g., "I wish I was never born") to specific thoughts/plans about ending one's life (e.g., "I thought of when I would kill myself"). The SIQ contains eight "critical" items (six on the SIQ Jr.) designated as such because they directly assess serious thoughts of self-destructive behavior: (1) "I thought about killing myself," (2) "I thought about how I would kill myself", (3) "I thought about when I would kill myself", (4) "I thought about what to write in a suicide note", (5) "I thought about writing a will", (6) "I thought about telling people I plan to kill myself", (7) "I thought about how easy it would be to end it all", and (8) "I thought if I had the chance I would kill myself". A participant with a total score of 41 or greater (31 or greater for the SIQ Jr.) is considered at risk for suicide. If a participant responds positively to 3 or more critical items (2 or more on SIQ Jr.), they are also considered to be positive for suicide risk on this measure[10].

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Demographic/Clinical Variables Questionnaire.—Participants completed a survey containing questions on demographic and clinical information (e.g. gender, age, diagnosis, reason for visit), history of medical or psychiatric illness, and prior healthcare usage, and opinions about suicide risk screening.

2.4 Procedure

We prospectively enrolled a convenience sample of 10- to 21-year-old patients from two different children's hospitals, November 2015 through July 2017. Exclusion criteria (assessed informally by parent/guardian (minors) or self-report (adult participants ≥18) included: (1) developmental disability, cognitive impairment, or communication disorder such that the patient was unable to comprehend questions/communicate answers; (2) medical illness suggesting that the patient was not physiologically stable enough to be approached; (3) parent/legal guardian of patients younger than 18 years unavailable for consent; and (4) non-English speaking parents/guardians and/or patients.

Site 1, Outpatient Specialty Clinics—Site 1 recruited patients who presented for an outpatient specialty visit to diabetes, endocrinology, sports medicine, or orthopedic clinics. Patients were approached for recruitment and enrolled upon check-in for their visit. Participants completed the study interview either before or after being seen by their provider. A trained research assistant (RA) with a minimum of a bachelor's degree verbally administered the ASQ, the SIQ/SIQ Jr, and other measures to participants, and responses were entered into REDCap[13] via tablet computers.

Site 2, Primary Care Clinic—Site 2 recruited patients who presented for an outpatient visit to primary care. Patients were approached for recruitment and enrolled upon check-in for their visit. Participants completed their primary care appointment with their provider and then the study interview began. A trained master's level research social worker (RSW) verbally administered the ASQ and the SIQ/SIQ Jr. to participants. Responses were recorded via paper instruments and subsequently entered into Microsoft Excel.

Sites 1 and 2 employed essentially the same procedures. To control for possible fatigue order effects, RA/RSWs alternated the order in which the ASQ and SIQ/SIQ Jr. were administered. In addition to the ASQ and SIQ/SIQ Jr., participants completed a demographic questionnaire that included questions about sociodemographic characteristics, history of medical and psychiatric illness, prior health care usage, and opinions about suicide risk screening. Interviews were conducted without the parent/guardian in the room, but participants under 18 years of age were told that if the RA/RSWs had any concerns about their safety, their parents would be notified, and pertinent information would be shared with the provider(s). As a safety measure, any participant who responded positively to any one of the 4 ASQ screening questions, or scored positive on the SIQ/SIQ Jr., or responded positively to any of the 8 SIQ/SIQ Jr. critical items required a further suicide risk assessment, which was conducted per standard of care for suicide risk at both study sites. Study participants were thanked for their participation, received a $10 gift card and were given a suicide prevention resource sheet that included facts, warning signs, and information about where to get help for suicidal ideation (including the National Suicide Prevention

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Lifeline and the Crisis Text Line) in addition to useful websites (e.g. American Foundation for Suicide Prevention, Suicide Prevention Resource Center).

2.5 Data Analysis

Sample size estimates were conducted using general sample size calculation for proportions[14]. We applied the central limit theorem with a sensitivity of 96.9%, prevalence rate of 15.8%[8,15,16] and a maximum acceptable margin of error d=3%. The total number of participants needed was 514.

A total of 918 patients were approached during the study period between the 2 sites; 827 patients (90.1%) were eligible for participation; 525 (response rate=63.5%) consented to participate. Seven patients had completed the consent/assent process and attended their outpatient specialty or primary care appointment, however did not complete the study interview for personal reasons (e.g. time) and another three participants were excluded due to RA/RSW error in conducting study interview or REDCap data entry and thus were excluded from analyses. All remaining participants (total n=515, comprised of 335 outpatient specialty clinic participants and 180 primary care participants) completed the screening protocol and were included in the analysis. (See Figure 1).

We examined the distributions of demographic characteristics and endorsement of four individual ASQ items by study site. Between-group differences (diabetes/endocrine v. sports medicine/orthopedic; outpatient specialty clinics v. primary care) were assessed using t-test for age (treated as a continuous variable), and chi-square tests or fisher's exact tests for categorical variables (sex/gender, race/ethnicity and insurance status). Traditional validity statistics were estimated in the package 'Epi'[17], a statistical analysis package used in epidemiology, including sensitivity, specificity, positive predictive value and the negative predictive value, LR +, LR −, area under curve (AUC) and the kappa statistic, where SIQ/SIQ Jr. used as 'true' test and 'ASQ' as 'screen' test[17,20]. Multivariate binary logistic regression modeling was used to examine the associations of suicide risk as detected by the ASQ (i.e., elevated risk present or absent) with the demographic, clinical and other variables (i.e., age, sex, race/ethnicity, insurance status, and site) entered into the model simultaneously. Adjusted odds ratios were the exponentiated values of multivariate logistic regression coefficients. Statistical significance was based on two-sided tests evaluated at the p=0.05 level of significance. Data preparation and data analyses were performed in R, version 1.1.442[18].

3. Results

Study participant characteristics are shown in Table 1. Approximately half of the participants (52.8%) were female, 39.6% were White, 19.1% were Black, and the mean age was 15.3 (*SD 2.7)* years. Most of participants (62.9%) reported having government-sponsored insurance. Compared to those from outpatient specialty sites, participants from the primary care site were more likely to be older, female, non-white and have private insurance. The indicators of the clinical validity of the ASQ, as measured by the concordance of the ASQ with the SIQ/SIQ Jr. in identifying individuals at elevated risk of suicide, are shown by study site in Table 2.

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Site 1 Outpatient Specialty Clinics

Forty-five (13.4%) of 335 participants screened positive for suicide risk on the ASQ. Of the participants who were positive on the ASQ, four participants were acute positive screens (1.2% of total sample). In the outpatient specialty clinics, the ASQ had a sensitivity of 100.0% (95% CI: 80.5 – 100.0%), specificity of 91.2% (95% CI: 87.5 – 94.1%), and negative predictive values at 100.0% (95% CI: 98.7–100.0). The positive LR was 11.4 (95% CI, 8.0–16.2), indicating that a positive screen on the ASQ was 11.4 times more likely to be seen in someone actually at suicidal risk than in someone not at risk. The 4-item ASQ screen had substantial agreement with the SIQ/SIQ Jr. for detecting suicide risk (AUC=95.6%)

Site 2 Primary Care Clinic

Twenty-eight (15.5%) of the 180 participants screened positive for suicide risk on the ASQ. All 28 positive screens were non-acute positive screens. In the primary care clinic, the ASQ had a sensitivity of 100.0% (95% CI: 59.0 – 100.0%), specificity of 87.9% (95% CI: 82.0 – 92.3%), and negative predictive values at 100.0% (95% CI: 97.7–100.0). The positive LR was 8.2 (95% CI, 7.7–13.1), indicating that a positive screen on the ASQ was 8.2 times more likely to be seen in someone actually at suicidal risk than in someone not at risk. The 4-item ASQ screen had substantial agreement with the SIQ/SIQ Jr. for detecting suicide risk (AUC=93.9%)

We examined the association between screening positive for suicide risk, as determined by the ASQ, and demographic variables (age, sex/gender, race, insurance status and site) in Table 3. For every one unit (year) increase in age, the odds of screening positive increased by 1.166 (95% CI: 1.036 – 1.316). Odds of screening positive for females were 2.358 (95% CI: 1.341 – 4.292) times higher compared to the odds for males. Odds of screening positive for publicly-insured were 0.540 (95% CI: 0.288 – 0.969) times higher compared to the odds for participants with private insurance.

4. Discussion

The ASQ appears to be a valid instrument for detecting elevated suicide risk in pediatric outpatient specialty or primary care clinics when compared to the standard criterion SIQ/SIQ Jr. This study demonstrated that the ASQ appeared to have good content validity. The four questions together assessed established suicide risk factors, including suicidal ideation, burdensomeness, and the most critical risk factor for future suicidal behavior—a previous suicide attempt. Positive responses to one or more of these four questions identified 100% of the youth at risk for suicide, as assessed by the SIQ/SIQ Jr., a much longer criterion standard proprietary instrument typically administered by mental health clinicians. Of note, the high specificity demonstrated by the four questions, reflecting the ASQ's ability to minimize false positives, or to efficiently and correctly identify youth who are currently *not* at risk for suicide (~91% outpatient specialty/~88% primary care), is important for a busy outpatient specialty and primary care clinic with limited and sometimes overburdened mental health resources. Given the consequences of failing to detect elevated risk for suicide, the high NPV (100.0% outpatient specialty/100% primary care), reflects the ability of the ASQ to avoid false negatives, or to correctly identify youth who *are* actually at elevated risk

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for suicide. Of note, the fairly high specificity demonstrated by the four questions, reflecting the ASQ's ability to efficiently and correctly identify youth who are currently *not* at risk for suicide (~91% outpatient specialty/~88% primary care), is important for a busy outpatient specialty and primary care clinic with limited and sometimes overburdened mental health resources. In contrast, the low PPV, or the proportion of positive screens on the ASQ that are also positive on the SIQ (~38% outpatient specialty/25% primary care), indicates that positive screens do not always reflect current severity of suicidal ideation (as assessed by the SIQ). This discordance is possibly attributable to the time parameter (lifetime) associated with item 4 of the ASQ, which assesses past suicide attempt (e.g. "Have you ever tried to kill yourself?). If a respondent has ever attempted suicide, they would screen positive on any subsequent ASQ screens, regardless of changes in suicidal ideation or actual suicide risk over time. Although a recent suicide attempt is certainly indicative of current acute suicide risk, one that occurred several years prior raises the patient's baseline risk for suicide but may not be very informative about current acute suicide risk. Therefore, patients who endorse Item 4 on the ASQ should be asked when the suicide attempt(s) occurred, and this information should be considered with respect to responses on Items 1–3, so that current risk can better be assessed.

Overall, 13.4% of the outpatient specialty clinic participants and 15.5% of the primary care participants screened positive for non-acute suicide risk (outpatient specialty n=45, primary care n=28), as assessed by the ASQ. Acute, or imminent risk for suicide was detected in only 4 participants in the outpatient specialty participants (1.2%) but was not detected for any participants in the primary care sample. This relatively small number of participants will likely not overburden a busy outpatient specialty or primary care clinic. Proactively identifying participants at elevated suicide risk with a brief screening instrument may protect a large number of youth in crisis.

The ASQ is intended to be utilized as a primary screening tool to identify patients who require further evaluation. Rational management of patients who screen positive can be conducted in a way that does not impede medical care, overwhelm the patient or overburden a busy outpatient pediatric practice. As found in this sample, the majority of patients screened will be negative and the ones that screen positive will be mostly non-acute positives in need of mental health linkage. It will be important to follow evidence-informed clinical pathways, such as the one developed by Brahmbhatt and colleagues, who recommend that non-acute positive screens be followed up with a brief suicide safety assessment to determine if any safety precautions or a more in-depth psychiatric evaluation are needed[32]. Conversely, acute positive screens, which are extremely rare among medical outpatients, as evidenced in this sample, do not require a brief suicide safety assessment because they are automatically considered an emergency and will require full safety precautions and full mental health/safety evaluation.

These results should be interpreted with consideration of a few limitations of this study. First, there are potential differences between our convenience sample and the general population of youth in non-emergency outpatient settings. As these differences would presumably impact participants' responses on the ASQ and the SIQ equally, they might affect the percent of patients that screen positive for suicide risk, but seem unlikely to impact

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the validation metrics, which are the focus of this study. One potential difference is that our sample was in urban, tertiary care academic hospitals and did not include other outpatient specialty and primary care clinics. However, we are aware of no a priori reason to think that the validation metrics we obtained for the ASQ would not generalize to other clinics. In addition,, we used convenience sampling, which may have introduced selection bias into our findings. As we were not able to collect data on non-participants, we could not assess this possibility statistically. If bias exists, it is likely that the percentage of non-study outpatient samples screening positive for suicide risk would be lower than with our convenience sample, but again, this potential difference likely would not affect the validation metrics.

Second, there may have been an effect of the timing of study participation (i.e. after their clinic appointment in primary care and either before or after their clinic appointment in outpatient specialty care) on the likelihood of screening positive for suicide risk. Participants may be more or less likely to report suicidality before or after an appointment. For example, appointment-related anticipatory anxiety and time pressure, or post-appointment fatigue, could theoretically affect results. In an actual clinical setting, patients would be screened before or during the appointment so that clinicians can review the responses. Data on the timing of the screen was not available, so we could not address the possibility of a timing effect. Again, if it is the case that timing of the screener influences the likelihood of endorsing suicide-related questions, presumably it affects the ASQ and the SIQ similarly, as they were administered at the same time (either before or after the appointment). Thus, this potential timing issue might affect the rate of positive screenings but probably not the validation metrics.

Third, there may have been a fatigue effect by asking the participants repeatedly about suicide ideation; however, any fatigue would have affected the two instruments equally as the order of the ASQ and SIQ/SIQ Jr. were rotated. Lastly, the tool used to validate the ASQ, the SIQ/SIQ-JR, is primarily intended to identify risk for suicide ideation and may not be predictive of suicide behavior.

5. Conclusion

The ASQ is a brief, valid and useful tool for screening for suicide risk in pediatric outpatient specialty and primary care clinics, which has been recommended by the Joint Commission and the American Academy of Pediatrics. The ASQ instrument and materials to guide implementation can be found in the ASQ Toolkit, a publicly available resource that can facilitate the detection of youth at risk for suicide (www.nimh.nih.gov/asq).

Future studies measuring the impact of suicide risk screening with youth in outpatient specialty and primary care clinics on outcomes such as linkage with mental health services and future suicidal behavior are warranted. Additionally, evaluating the acceptance of such screening instruments by clinicians and the costs associated with implementing universal suicide risk screening in outpatient specialty and primary care clinics would inform ASQ implementation strategies. A secondary exploration of the ASQ on an item level would provide nuanced information regarding the endorsement of individual items for the same subject and between subjects, allowing us to better understand risk for suicide within

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different hospital settings. Moreover, it would be useful to examine the utility of the ASQ in conjunction with brief suicide interventions similar to the Screening and Brief Intervention to Treatment model[19], given the push for integrated behavioral health in these settings.

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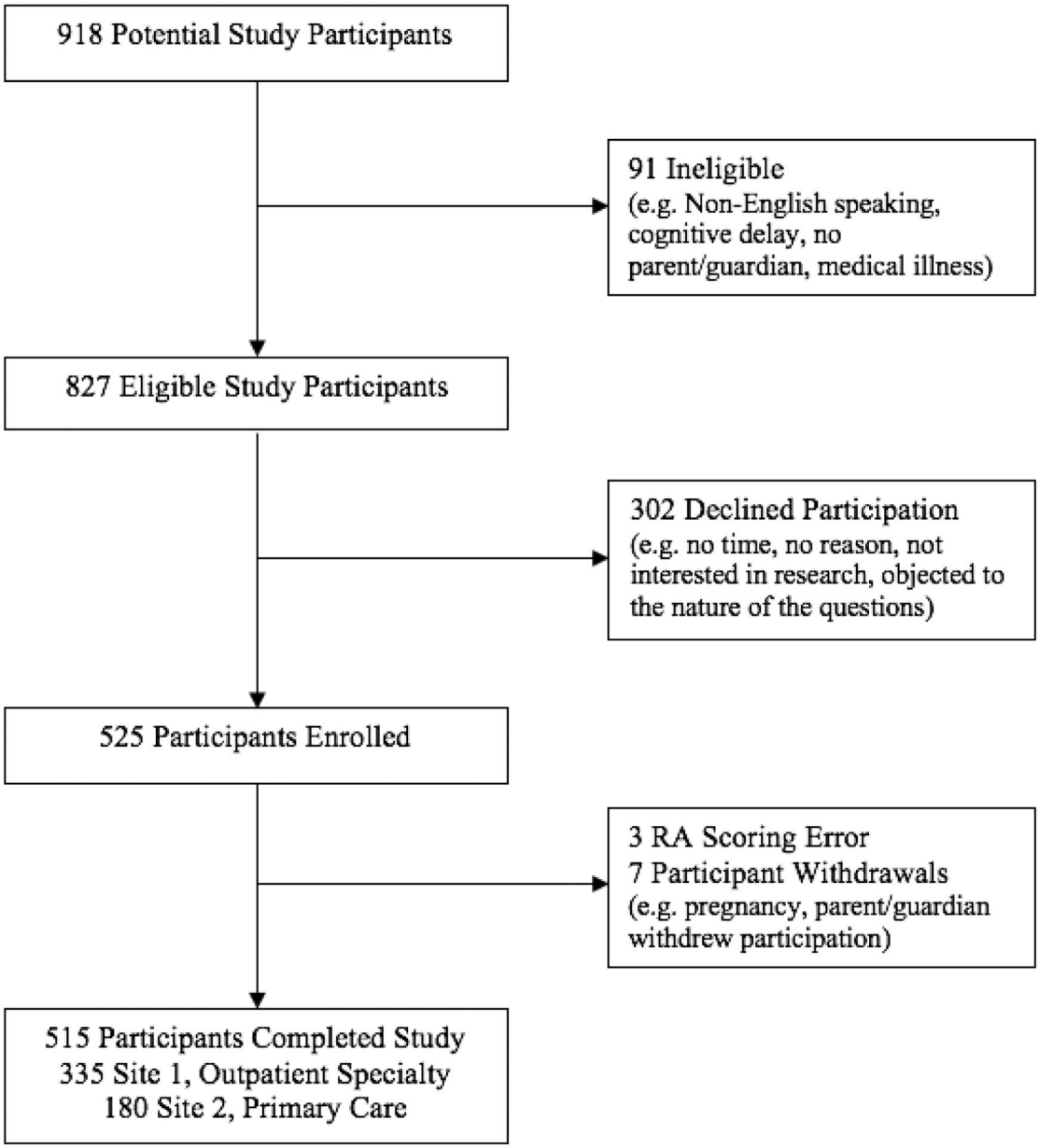

Figure 1.
Sample Flowchart

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Table 1.

Sample characteristics of clinic settings and study sites

Characteristics		No. (%)					
		Site 1			Site 2	Tests for Group Differences:	
		Outpatient Specialty				(1) vs. (2)	(3) vs. (4)
		Diabetes/ Endocrine	Sports Med/Ortho	Subtotal	Primary Care	Dia/Endo vs. SportsOrtho	Outpatient Specialty vs. Primary Care
	Total Sample	(1)	(2)	(3)	(4)		p-values
Sample Size	515	197	138	335	180		
Age in years, MEAN (SD)	15.3 (2.7)	14.6 (2.7)	14.0 (2.1)	14.4 (2.4)	17.2 (2.1)	0.042 *	<.0001 *
Sex						0.029 *	0.233
Female	272 (52.8)	107 (54.3)	60 (43.5)	167 (49.9)	105 (58.3)		
Male	231 (44.9)	80 (40.6)	76 (55.1)	156 (46.6)	75 (41.7)		
Transgender	9 (1.7)	9 (4.6)	0 (0.0)	9 (2.7)	0 (0.0)		
Unknown	3 (.6)	1 (.5)	2 (1.4)	3 (.9)	0 (0.0)		
Race/ethnicity						0.383	<.0001 *
White	258 (50.1)	132 (67.0)	98 (71.0)	230 (68.7)	28 (15.6)		
Black	95 (18.4)	22 (11.2)	15 (10.9)	37 (11.0)	58 (32.2)		
Hispanic/Latino	105 (20.4)	25 (12.7)	14 (10.1)	39 (11.6)	66 (36.7)		
Asian	36 (7.0)	11 (5.6)	2 (1.4)	13 (3.9)	23 (12.8)		
Other/Unknown	21 (4.1)	7 (3.6)	9 (6.5)	16 (4.8)	5 (2.8)		
Insurance						0.785	0.726
Private	324 (62.9)	123 (62.4)	85 (61.6)	208 (62.1)	116 (64.4)		
Public	170 (33.0)	66 (33.5)	40 (29.0)	106 (31.6)	64 (35.6)		
None/Unknown	21 (4.1)	8 (4.1)	13 (9.4)	21 (6.3)	0 (0.0)		

Note:

""* $p \leq 0.05$,

*"***"* $p \leq 0.001$

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Table 2.

Concordance of SIQ/SIQ Jr. and suicide risk assessed by ASQ for total sample and by study site

Suicide Risk	Sample Size	ASQ (Screen)	SIQ/SIQ Jr. (True)	Sensitivity	Specificity	PPV	NPV	LR+	LR−	Kappa	AUC
		Positive (n, %)	Positive (n, %)	% (95% CI)	% (95% CI)	% (95% CI)	% (95% CI)	% (95% CI)	% (95% CI)		
Site I: Outpatient Specialty	335	45 (13.4)	17 (5.0)	100.0 (80.5–100.0)	91.2 (87.5–94.1)	37.8 (23.8–53.5)	100.0 (98.7–100.0)	11.4 (8.0–16.2)	0.0 (0.0–NaN)	0.512	0.956
Site II: Primary Care	180	28 (15.5)	7 (3.9)	100 (59.0–1)	87.9 (82.0–92.3)	25.0 (10.7–44.9)	100.0 (97.7–100.0)	8.2 (5.5–12.3)	0.0 (0.0–NaN)	0.360	0.939

Note: NaN= Not a number[20]

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Table 3.

The associations of risk for suicide as detected by the ASQ and demographic characteristics

Variable	OR	95% CI		p-value
		Lower	Upper	
Site				
Primary Care	Reference			
Outpatient	1.117	0.544	2.293	0.762
Age	1.166	1.036	1.316	.012 *
Sex				
Male	Reference			
Female	2.358	1.341	4.292	0.004 **
Race				
White	Reference			
Black	0.525	0.206	1.231	0.154
Other/U nknown	0.979	0.49	1.925	0.951
Insurance				
Private	Reference			
Public	0.540	0.288	0.969	0.045 *

Note: "." Denotes .1 significance (trending towards),

""* Denotes ≤0.05 significance,

*"**"* Denotes ≤0.01 significance,

*"***"* Denotes ≤0.001 significance